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CSA5115-Crytptography and Network Security
QR Code-Based Secure Authentication System

Presented by:

Elakati Renu Sree(192511035)

Tarunika.S(192511024)

Sweta.R(192511408)

Abinaya.R(192421161)

Guided by:

Dr. T. Sangeetha

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ABSTRACT

A QR Code-Based Secure Authentication System is a modern way to verify a user's identity using QR codes. Instead of typing passwords every time, users can scan a QR code with their mobile phone to log in quickly and securely. The system generates a unique QR code for each login request, making it difficult for attackers to steal or reuse it. This method improves security, saves time, and provides a user-friendly login experience. It is widely used in banking, online services, educational institutions, and office access systems.

Introduction

- QR Code-Based Secure Authentication System is a secure method for user login and identity verification.
- It uses QR codes to provide a fast and easy authentication process.
- User details are encrypted and stored securely inside the QR code.
- The system scans and verifies the QR code before allowing access.
- It checks the validity and expiry of the QR code to prevent misuse.
- Successful and failed login attempts are recorded in an activity log.
- This system helps reduce risks such as password theft, unauthorized access, and fake login attempts.
- It can be used in websites, mobile applications, offices, colleges, and secure systems.

Problem Statement

- **Password Vulnerability** – Traditional passwords can be stolen, guessed, or shared.
- **Unauthorized Access** – Attackers may use stolen credentials to access the system.
- **Need for Secure Login** – A faster and safer authentication method is required.
- **QR Code Verification** – Encrypted QR codes can be used to securely verify user identity.
- **Activity Monitoring** – The system should record successful and failed login attempts for better security

Objectives

- Provide secure user authentication using QR codes.
- Protect user information using encryption.
- Prevent unauthorized access to the system.
- Verify QR code validity and expiry.
- Maintain login and activity logs for security monitoring.
- Provide a fast and user-friendly login process.
- Reduce the risk of password theft and misuse.

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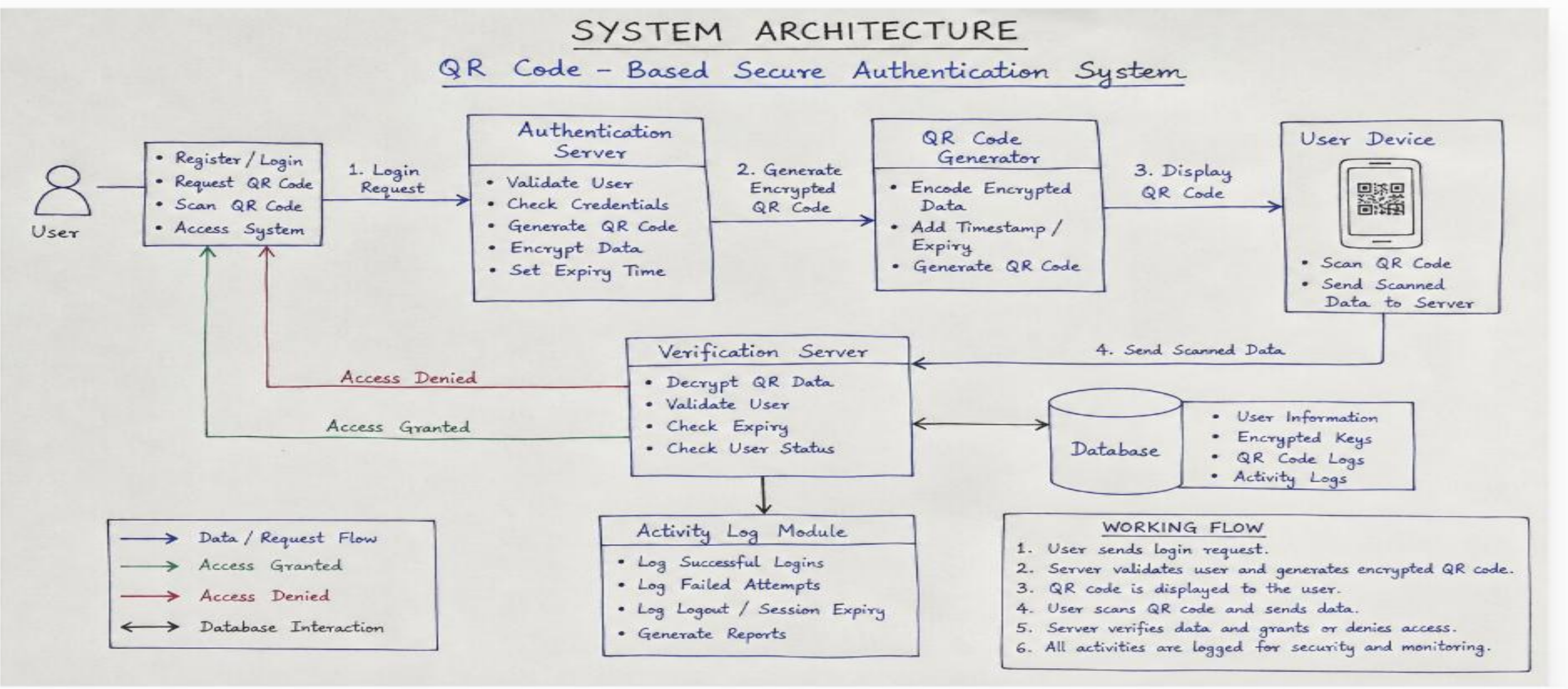
LITERATURE SURVEY

S.No	Author(s) & Year	Title of the Paper	Methodology / Technology Used	Findings	Research Gap
1	M. Kumar et al., 2024	QR Code-Based Authentication for Secure Access	Used QR code generation and scanning with encrypted user credentials	Provided fast and convenient user authentication	Limited protection against QR code copying and replay attacks
2	S. Patel and R. Sharma, 2024	Secure QR Code Authentication System Using Encryption	Applied encryption techniques to protect authentication data stored in QR codes	Improved confidentiality and reduced unauthorized access	Requires stronger mechanisms for key management and QR validation
3	A. Zhang et al., 2025	Dynamic QR Code-Based User Authentication	Generated time-based dynamic QR codes for user verification	Reduced the risk of QR code reuse and replay attacks	Dynamic code generation may require reliable time synchronization
4	P. Singh et al., 2025	Encrypted QR Authentication for Mobile and Web Applications	Combined QR scanning, encryption, and server-side authentication	Improved security and provided a user-friendly login process	Limited integration with activity monitoring and failed-login detection
5	R. Chen et al., 2026	Multi-Factor Authentication Using QR Codes	Combined QR authentication with passwords/OTP and secure verification	Increased authentication security by using multiple verification methods	Higher implementation complexity and additional verification time

System architecture



Module 1: User Registration

- User enters personal details (name, email, phone number, password).
- System validates the entered information.
- Password is securely stored in encrypted form.
- A unique user account is created.
- User can access the system after successful registration.

Module 2: QR Code Generation

- System generates a unique QR code for each login request.
- QR code contains an encrypted authentication token.
- QR code is displayed on the login screen.
- Each QR code is valid for a limited time.
- Prevents unauthorized reuse of QR codes.

Module 3: Encrypted QR Verification

- User scans the QR code using a mobile device.
- The system decrypts the QR code data.
- Verifies the authenticity of the QR code.
- Checks whether the QR code has expired or has already been used.
- Allows only valid QR codes for authentication.

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Module 4: Login Authentication & Activity Log

- Grants access only after successful authentication.
- Records every login and logout activity.
- Stores the date, time, and login status (successful or failed).
- Prevents unauthorized access to the system.
- Maintains an activity log for security monitoring and future reference.

Tools/Technologies used

- Language: Python 3.x
- Framework: Flask
- Database: SQLite3 (or MySQL)
- QR Code Generation: qrcode
- QR Code Scanning: pyzbar
- Image Processing: OpenCV (cv2)
- Encryption: cryptography (Fernet)
- Password Security: bcrypt
- IDE: Visual Studio Code

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Result

- Successfully implemented QR code-based user authentication.
- Securely generated encrypted QR codes for user verification.
- Successfully verified valid and expired QR codes.
- Prevented unauthorized users from accessing the system.
- Recorded successful and failed login attempts in activity logs.
- Provided a fast and user-friendly authentication process.
- Improved overall security and reliability compared with traditional password-based login.

Conclusion

- The system provides a secure and fast authentication method using QR codes.
- Encryption protects sensitive user information from unauthorized access.
- QR verification and expiry checking help prevent misuse and repeated attacks.
- Activity logs help monitor successful and failed authentication attempts.
- The system reduces dependence on traditional password-based authentication.
- Overall, it provides a reliable, user-friendly, and secure authentication solution.

IEEE Standards

- **IEEE 802.1X** – Secure network authentication and access control.
- **ISO/IEC 18004:2024** – QR code generation and decoding standard.
- **NIST SP 800-63B-4** – Secure digital authentication guidelines.
- **Encryption & Protected Communication** – Protects authentication data during transmission.
- **Access Control & Logging** – Prevents unauthorized access and maintains authentication records.

Future scope

- Add fingerprint and face recognition for extra security.
- Use OTP with QR codes for multi-factor authentication.
- Generate dynamic QR codes that expire quickly.
- Develop a mobile app for easy authentication.
- Use cloud storage for secure data management.
- Add AI-based security to detect suspicious login attempts.
- Improve the system for use in colleges, offices, banks, and other organizations.

References

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- RFC 8446, The Transport Layer Security (TLS) Protocol Version 1.3, Internet Engineering Task Force (IETF), 2018.
- OWASP, Authentication Cheat Sheet, Open Web Application Security Project.
- OWASP, Cryptographic Storage Cheat Sheet, Open Web Application Security Project.
- W3C Web Authentication (WebAuthn), Web Authentication API, World Wide Web Consortium

Thank you